Scientific Research Report

A Multimodal Model for Caries Screening Using Intraoral Images and Questionnaires

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ARTICLE INFO

Article history:
Received 29 October 2025
Received in revised form
7 January 2026
Accepted 9 January 2026
Available online 5 February 2026

Key words:
Artificial intelligence
Caries detection
Dental public health
Preventive dentistry
Paediatric caries

ABSTRACT

Introduction and aims: To develop a multimodal deep learning model for dental caries screening in children by integrating intraoral photographs and questionnaire-based data, and to compare its performance against that of a unimodal model utilising images alone.

Methods: In total, 7671 images (3913 occlusal and 3758 smooth surfaces) were collected from children across different dentition stages. Oral health questionnaires were also conducted. The images were clinically labelled into three categories: caries-free, early caries, and moderate-to-severe caries. A multimodal model integrating intraoral images and questionnaire data was trained and evaluated against an image-only unimodal model. The performance metrics included accuracy, precision, recall, F1 score, and the area under the receiver operating characteristic curve (ROC-AUC). SHapley Additive exPlanations (SHAP) analysis was conducted to interpret feature importance.

Results: For the multimodal model, occlusal surface classification achieved an accuracy of 92.3% for the caries-free category, 89.2% for the early caries category, and 93.7% for the moderate-to-severe caries category, with ROC-AUCs of 0.967, 0.911, and 0.972, respectively. The performance of smooth surfaces was similarly strong (an accuracy of 92.1% for the caries-free category, 96.7% for the early caries category, and 93.4% for the moderate-to-severe caries category), with ROC-AUCs of 0.938, 0.908, and 0.963, respectively. Incorporating questionnaire data improved recall for the early caries category without compromising overall accuracy. SHAP analysis provided interpretable insights into model decision-making.

Conclusion: Compared with the image-only unimodal model, the multimodal deep learning model increased early caries recall while preserving overall diagnostic performance.

Clinical relevance: This study describes a multimodal caries intelligent screening model, demonstrating its potential as a population-level screening tool. However, further large-scale, multicentre validation is necessary to confirm its generalisability and effectiveness.

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Introduction

Dental caries remains highly prevalent among children.[1] According to the Fourth National Oral Health Survey in China, the prevalence of dental caries was 71.9% in deciduous teeth among 5-year-olds and 38.5% in permanent teeth among 12-year-olds.[2] Moreover, global oral health care resources remain insufficient and inequitably distributed. For example, fluoride toothpaste is recognised as essential for preventing dental caries and is included in the WHO Model List of Essential

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https://doi.org/10.1016/j.identj.2026.109420
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Medicines; however, it remains inaccessible to many patients in low-income settings.[4] The global oral health workforce is highly inequitable and marked by significant disparities in the density of dental personnel between high-income and low-income countries, as well as between urban and rural areas.[5,6] This scarcity hinders large-scale oral health screening programs and limits early intervention efforts, presenting a major barrier to improving paediatric oral health.

Visual examination is clinically preferred for the detection of caries as it is minimally technique sensitive and provides diagnostically sufficient accuracy after tooth cleaning and drying.[7-9] Even for the early diagnosis of caries, it has moderate to high sensitivity and specificity.[3] This conventional visual assessment is being increasingly supported by artificial intelligence (AI). Compared with radiographs, intraoral images offer distinct practical advantages in real-world screening, such as portability, cost-effectiveness, and suitability for school-based programs. Recent studies have explored the application of AI-driven automated systems for caries detection using intraoral images and have demonstrated promising diagnostic performance.[10-13]

AI, particularly deep learning techniques exemplified by convolutional neural networks (CNNs), has become a vital tool for image recognition and analysis in dentistry, enabling the automated extraction of relevant features from medical images such as intraoral photographs, radiographs, and cone-beam CT scans.[10-13] Unlike large language models such as ChatGPT, which focus on language generation,[14] CNNs excel in object detection and classification. By learning directly from raw pixel data without requiring manual feature engineering, CNNs can effectively classify images. Region-based convolutional neural networks (R-CNNs) were specifically designed for object detection, enabling the automatic identification and annotation of target objects (regions of interest).[15,16] This architecture was subsequently refined into the more efficient Faster R-CNN and further extended into the Mask R-CNN method, which not only detects targets within images but also delivers high-quality pixel-level segmentation results.[17,18] In recent years, Mask R-CNN has been applied across various fields, including object extraction,[19] and has shown utility in dental medicine for assisting in the localisation of causative teeth.[20]

Moreover, the YOLO series of algorithms, known for real-time detection, has made significant progress in caries detection. The YOLO model proposed by Redmon et al[21] in 2016 demonstrated excellent performance in real-time detection tasks across thousands of classes. The YOLOv3 version released in 2018 achieved an accuracy of 87.4% in detecting early caries lesions using intraoral photographs taken with smartphones, significantly outperforming the contemporaneous Faster R-CNN at 71.4%.[22] The architecture has since evolved through versions from YOLOv4 to YOLOv11, incorporating attention mechanisms and multiscale detection to better localise subtle lesions under complex conditions.[23] Notably, the accuracy of YOLOv7 was improved by the introduction of the 'bag of freebies' training strategy without increasing the computational cost,[24-26] making it particularly advantageous for detecting minute carious lesions in intraoral radiographs and providing robust technical support for early caries diagnosis.

Dental caries is a dynamic disease process that, when identified in its initial stages, can be arrested and even reversed through targeted interventions.[27,28] Early caries screening enables timely detection of these hidden lesions and allows for noninvasive interventions, such as dietary counselling, plaque control, fluoride products, and pit and fissure sealants.[26] This approach effectively arrests the progression of caries, significantly reduces the number of advanced cases requiring invasive restoration, and helps control health care costs at the population level. Therefore, early caries screening is a key strategy for shifting the focus of caries management upstream, successfully bridging the gap between 'prevention' and 'minimally invasive treatment'. Despite extensive efforts in exploring various methodologies, the performance of AI in detecting early caries has yet to be optimised.[29-31]

Current AI models for caries diagnosis predominantly rely on imaging features,[32] largely neglecting crucial behavioural determinants of caries risk, such as dietary habits and oral hygiene practices. While epidemiological studies have firmly established the impact of behavioural and environmental factors on the development and progression of dental caries,[33] the question of how to effectively quantify these factors and combine them with imaging features to improve AI-based caries detection, particularly in early stages, remains an area that has received limited research attention. Multimodality models enable AI systems to process and interpret information from diverse modalities, such as text, images, and voice, facilitating comprehensive patient data analysis and demonstrating significant potential in the field of dentistry.[34] To the best of our knowledge, no AI-based caries screening model utilising multimodal data has been reported. Therefore, this study aims to develop a multimodal deep learning model for dental caries screening in children by integrating intraoral photographs and questionnaire-based data and to compare its performance to that of a unimodal model that utilises images alone. Kühnisch et al[11] constructed a CNN model based on standardised intraoral photographs, achieving an accuracy of 92.5% in diagnosing dental caries. Building on this, we hypothesise that integrating questionnaire-derived data on behavioural factors will increase the diagnostic accuracy of caries, especially for early caries, with an anticipated performance target of at least 90%.

Methods

This study was reported following the recommendations of the Checklist for Artificial Intelligence in Medical Imaging (CLAIM). The study flowchart is presented in Figure 1.

Sample size calculation

In this cross-sectional study, intraoral images and oral health-related questionnaires were collected from children in Zengcheng District, Guangzhou, who were stratified into three age groups (3-6, 7-9, and 12-15 years). The sample size was calculated using the ROC-AUC confidence interval module in PASS (v15.0.5). It was based on an anticipated AUC of 0.90, a confidence interval width of 0.196, a confidence level

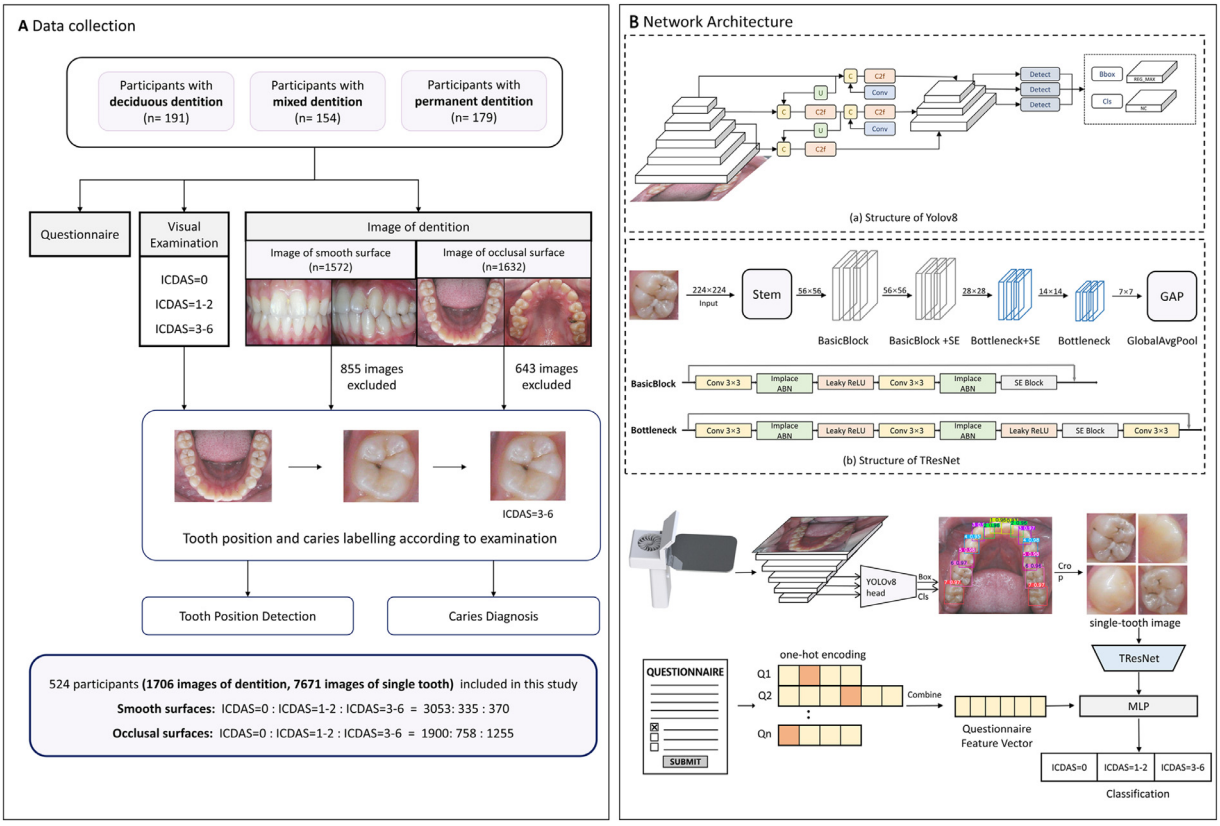


Fig. 1 – Overview of this study. A, data collection process. B, construction of the multimodal model for caries classification. ABN, In-Place Activated BatchNorm; Conv, convolution; ReLU, rectified linear unit; SE block, squeeze-and-excitation block.

of 95% and an α = 0.05. Due to the limited published data on the International Caries Detection and Assessment System (ICDAS) score distribution in these populations, we estimated prevalence rates according to the results of the Fourth National Oral Health Survey of China, using WHO-standard caries detection rates as a proxy for ICDAS score of 3 to 6.[35] For ICDAS scores of 0 and 1 to 2, which lack prevalence data, a conservative 1:1 distribution was assumed to prevent category imbalance. Sample size calculations indicated that a minimum sample size of 129 participants would provide sufficient statistical power for diagnostic comparisons across the different caries categories. However, as the aim of this study was to develop a real-world screening model, ensuring model generalisability and robustness was paramount. Thus, a larger cohort is beneficial for capturing data variability, reducing overfitting, and enabling detailed analyses. Given practical and experimental personnel constraints, we ultimately enrolled 524 participants to enhance the generalisability and robustness of the model within the school-based real-world setting.

Intraoral image data acquisition

A self-developed standardised intraoral imaging device was used to capture intraoral photographs in an indoor setting. The standardised device consists primarily of the following components: a housing (Tuozhu Technology Company), a turbine fan (JinGuang Dental Company), a camera module (Sony Group Corporation), a grip handle (Tuozhu Technology Company), an LED light (Jialichuang Technology Company) and a metal reflector (JinGuang Dental Company). The structure of the self-developed standardised intraoral imaging devices is provided in Appendix 1. The housing serves as the main structural framework of the device. The built-in turbine fan generates airflow to disperse moisture from the breath, thereby preventing the reflector from fogging due to condensation. The camera module is equipped with a Sony IMX258 image sensor and an adjustable LED lighting unit, enabling the capture of high-resolution, clear images. The handle features two independent buttons for separately controlling the fan and the light source. The device uses a removable and replaceable standard metal intraoral mirror as the reflective component. The angle and distance between the camera and the reflector were preconfigured based on typical dental arch anatomy, average mouth opening range, and clinical practice. This design allows nonprofessionals to acquire appropriately angled intraoral images without the need for additional adjustments during use. The device connects to a computer via a USB 2.0 interface. The computer displays the real-time video feed and controls focus and image capture. By firmly placing the metal reflector against the occlusal plane (cusp tips) of either the maxillary or mandibular dental arch, the operator can obtain intraoral images with optimal brightness and viewing angles.

To accommodate the anatomical characteristics of children in different age groups, we implemented an age-adapted acquisition protocol: for children aged 3 to 6 and 7 to 9 years, a small reflector was used, which achieves autofocus and

imaging upon gentle contact with the tooth surface; for participants aged 12 to 15 years, a standard reflector was used. To ensure infection control, an individually sterilised reflector was used for each participant and replaced after every use. Before the formal study, we conducted a randomised controlled trial to compare the image quality obtained with this device against that captured by a smartphone. The results demonstrated that the intraoral images acquired with this standardised device were significantly superior in quality to those taken with a smartphone (the relevant data have been compiled and submitted to another journal and are therefore not presented in this article).

The setup permitted image acquisition with patients remaining seated throughout the procedure. Paediatric participants were instructed to fully expose their teeth and gingival surfaces while maintaining a dry examination field, achieved through the use of gauze and cotton rolls for saliva control. The imaging protocol included photographs of the maxillary arch, complete mandibular arch, and labial and buccal surfaces of the teeth in the intercuspal position. Standardised photographs of the labial/buccal surfaces and the occlusal surface of each tooth were obtained in this study. Notably, while separate images were not acquired specifically for the palatal surfaces of maxillary posterior teeth or the lingual surfaces of mandibular posterior teeth, these surfaces were naturally exposed and captured together with the occlusal surface in the same occlusal-view image. Thus, all tooth surfaces visible in the views were included in the model's comprehensive analysis for training and evaluation. In cases where a single image could not adequately capture the entire arch, multiple overlapping photographs were obtained to ensure comprehensive documentation of all tooth surfaces. All images, including those of the labial/buccal and occlusal surfaces, were saved in uncompressed JPG format at a resolution of 3840 × 3140 pixels (96 dpi) to ensure high-quality archival data and for subsequent analysis.

Questionnaire

The questionnaire used in this study was adapted from the official questionnaire of the Fourth National Oral Health Epidemiological Survey of China.[2,36] The only modification made to the original questionnaire was the removal of the 'household income' item. No other changes were introduced to the content or structure of the questionnaire. During the pilot phase, we found that the 'household income' item was widely perceived as sensitive by parents, leading to a noticeable decline in willingness to respond. To improve questionnaire completion rates and data quality and to avoid selective nonresponse due to sensitive questions, we omitted this item.

The questionnaire (Appendix 2) encompasses five key domains: (1) children's dietary habits, (2) oral hygiene practices, (3) the utilisation of oral health services, (4) oral health knowledge and attitudes among children and their parents, and (5) oral health-related quality of life (OHRQoL). For participants aged 3 to 6 years and 7 to 9 years, questionnaires were administered through group instruction followed by parent-assisted completion. After receiving standardised group instructions, the children in the 12 to 15-year age group completed the questionnaires independently. All of the collected data were anonymised through the removal of personally identifiable information before analysis to ensure participant confidentiality.

Caries examination and image labelling

Caries examinations were performed by an experienced examiner (LY. P, >10 years of clinical practice). The examiner evaluated the caries according to the International Caries Detection and Assessment System II (ICDAS-II) criteria[37] and recorded the findings for subsequent image labelling. Caries were categorised as follows: ICDAS score 0 = caries-free; scores 1 to 2 = early caries; and scores ≥3 = moderate-to-severe caries.[38] To evaluate intra-examiner reliability, repeated examinations were performed on 10% of the participants after the initial clinical examination. The calculated kappa values were 0.90 for caries-free, 0.78 for early caries, and 0.98 for moderate-to-severe caries, indicating that the examiners' assessments demonstrated good reliability.

Before annotation, all images underwent rigorous quality assessment. The exclusion criteria were based on two main considerations. First, for technical quality, we excluded images with any of the following defects: duplication, focal blurring, exposure abnormalities (over or underexposure), or obstruction by foreign objects.[11,39] Second, to ensure consistency with the study objectives and to minimise potential bias, we included only images of either healthy tooth surfaces or surfaces with visually detectable caries. All images showing noncarious hard-tissue conditions, such as enamel hypomineralisation, hypoplasia, erosion, tooth wear, fissure sealants, or direct/indirect restorations, were excluded. Furthermore, images corresponding to early lesions that were noted during clinical examinations but were not clearly visible or identifiable in the photographs, such as early approximal caries at the cervical area or early caries on cusp tips, were also excluded.

Dental arch images were systematically processed to isolate individual teeth. The annotation protocol used a two-stage process: primary annotation of tooth positional markers and subsequent caries classification. Another experienced dentist (ZY, L) used LabelMe software (v5.1.1) to annotate the intraoral photographs according to the clinical examination records to create a labelled dataset for model development. All data were anonymised and stored securely for analysis.

Model training and testing

This study used a three-stage modelling framework for automated caries detection and classification: tooth position detection, image-based (unimodal) caries classification, and multimodal caries classification, which integrates intraoral images with questionnaire-derived oral health data. Here, the 'unimodal' model refers specifically to a caries screening model trained solely on intraoral photographs, whereas the 'multimodal' model refers to a caries screening model trained by integrating both intraoral photographs and questionnaire information (such as oral hygiene habits and dietary habits).

Stage 1: Tooth position detection. The collected intraoral images were processed using a YOLOv8-based detection model to localise, segment, and annotate individual teeth. This stage generated high-resolution cropped images of single teeth, guided by YOLO-predicted positional annotations, to serve as input for subsequent caries classification. Stage 2: Image-based caries classification. High-resolution single-tooth images were classified into three clinically validated categories: caries-free, early caries, or moderate-to-severe caries. To optimise feature extraction for subtle caries lesions, we implemented a TResNet architecture, an enhanced variant of ResNet with superior discriminative capabilities.[30] Given the divergent morphological features of caries on smooth and occlusal surfaces, separate models were trained for each surface type. Stage 3: Multimodal caries classification. Image features extracted via TResNet were fused with one-hot encoded oral health questionnaire data. The combined features were processed through a multilayer perceptron (MLP) to generate the final classification.

The data were partitioned in this study at the participant level. First, data from 20% of the participants were reserved as a completely independent test set, which did not participate in any training or model selection processes. The remaining 80% of the data were utilised for model development, with 64% allocated as the training set and 16% as the validation set for hyperparameter tuning. After the optimal combination of hyperparameters was determined, the training and validation sets were merged to retrain the model. The model weights obtained from the final training epoch were selected as the final model, whose performance was ultimately evaluated on the reserved independent test set. To enhance robustness, we applied extensive data augmentation, including random adjustments to brightness, saturation, rotation, translation, scaling, and cropping, followed by normalisation. Input resolutions were standardised at 384 × 384 for full-arch images (Stage 1) and 224 × 224 for single-tooth images (Stages 2-3).

Statistical analysis

The data were analysed using Python (http://www.python.org, version 3.11.5). The detection performance of tooth positions was evaluated using standard metrics, including accuracy, precision, recall, F1 score, and ROC-AUC. For the tooth position detection model, global performance was assessed using the mean average precision at an intersection-over-union threshold of 50% (mAP50) and across thresholds ranging from 50% to 95% (mAP50-95). The caries classification model was evaluated based on overall accuracy, accuracy, precision, recall, F1 score and ROC-AUC. Within each dentition subgroup (primary, mixed, and permanent dentition stages), model performance was evaluated using the macro-averaged F1 scores and class-specific F1 scores. Differences in performance between dentition subgroups were assessed using a bootstrap-based pairwise comparison approach. The mean difference and its 95% confidence interval were estimated using the percentile method. To enhance the interpretability of questionnaire-derived features, Shapley Additive exPlanations (SHAP) analysis was used.

Results

The tooth position detection model demonstrated consistently high accuracy (>98.5%) across all tooth positions (Table 1) and achieved outstanding overall performance, with a mean average precision at an IoU 50% (mAP50) of 0.96 and an mAP50-95 of 0.937.

Both the multimodal and unimodal caries classification models exhibited strong diagnostic performance across all caries categories (Table 2). On occlusal surfaces, the unimodal model achieved accuracies of 91.5% for healthy teeth, 90.0% for early caries, and 93.5% for moderate-to-severe caries, with ROC-AUC values of 0.970, 0.929, and 0.968, respectively. The multimodal model showed slightly improved accuracy in identifying healthy teeth and maintained comparable performance for moderate-to-severe caries. Although a marginal decrease in early caries accuracy was observed in the multimodal model, recall improved substantially (multimodal: 0.732 vs unimodal: 0.673).

Similar trends were observed on smooth surfaces. The unimodal model attained accuracies of 92.4% for sound surfaces, 93.6% for early caries, and 97.1% for moderate-to-severe caries, with ROC-AUC values of 0.945, 0.912, and 0.968, respectively. The multimodal model maintained diagnostic accuracy across all categories but significantly improved recall for early caries detection (multimodal: 0.677 vs unimodal: 0.581).

Table 3 compares the performance of multimodal and unimodal models for caries across different dentition stages

Table 1 – Performance of the tooth position detection model.

Tooth position	Accuracy	Precision	Recall	F1-score	ROC-AUC
Deciduous central incisor	0.995	0.971	0.936	0.953	0.950
Deciduous lateral incisors	0.985	0.924	0.868	0.895	0.939
Deciduous canine	0.985	0.944	0.935	0.939	0.982
The first deciduous molar	0.994	0.982	0.968	0.975	0.986
The second deciduous molar	0.993	0.975	0.962	0.969	0.985
Central incisor	0.994	0.972	0.975	0.974	0.982
Lateral incisor	0.985	0.898	0.946	0.922	0.965
Canine	0.987	0.847	0.877	0.862	0.929
The first premolar	0.995	0.937	0.971	0.954	0.986
The second premolar	0.997	0.969	0.977	0.973	0.977
The first permanent molar	0.993	0.968	0.972	0.970	0.969
The second permanent molar	0.998	0.960	0.990	0.974	0.992

Table 2 – Comparison of model performance between multimodal and unimodal models.

Tooth surface	Detection of caries	Accuracy		Precision		Recall		F1-score		ROC-AUC	
		Multimodal	Unimodal	Multimodal	Unimodal	Multimodal	Unimodal	Multimodal	Unimodal	Multimodal	Unimodal
Occlusal surface	Caries free	0.923	0.915	0.915	0.919	0.923	0.901	0.919	0.898	0.967	0.970
	Early caries	0.892	0.900	0.727	0.792	0.732	0.673	0.730	0.728	0.911	0.929
	Moderate-to-severe caries	0.937	0.935	0.947	0.882	0.857	0.924	0.900	0.903	0.972	0.968
	Overall performance	0.917	0.917	0.863	0.864	0.837	0.833	0.849	0.847	0.960	0.961
Smooth surface	Caries free	0.921	0.924	0.940	0.947	0.966	0.961	0.953	0.954	0.938	0.945
	Early caries	0.934	0.936	0.583	0.621	0.677	0.581	0.627	0.600	0.908	0.912
	Moderate-to-severe caries	0.967	0.971	0.878	0.870	0.694	0.758	0.775	0.810	0.963	0.968
	Overall performance	0.941	0.944	0.800	0.813	0.779	0.767	0.785	0.788	0.966	0.980

Table 3 – Subgroup analysis across different dentition stages.

Dataset	Model	Category	F1-score			Performance differences between dentition		
			Primary	Mixed	Permanent	Δ(Permanent-primary)	Δ(Mixed-primary)	Δ(Permanent-mixed)
Occlusal Surface	Unimodal	Macro	0.833	0.843	0.848	(0.015, [−0.060, 0.095])	(0.010, [−0.063, 0.088])	(0.005, [−0.062, 0.071])
		Caries free	0.899	0.898	0.936	(0.037, [−0.017, 0.096])	(−0.001, [−0.060, 0.064])	(0.038, [−0.010, 0.088])
		Early caries	0.675	0.741	0.737	(0.062, [−0.103, 0.230])	(0.066, [−0.085, 0.232])	(−0.004, [−0.130, 0.121])
		Moderate-to-severe caries	0.925	0.891	0.872	(−0.053, [−0.123, 0.016])	(−0.034, [−0.097, 0.028])	(−0.020, [−0.096, 0.055])
	Multimodal	Macro	0.874	0.886	0.789	(−0.084, [−0.162, −0.007])*	(0.013, [−0.049, 0.079])	(−0.097, [−0.165, −0.031])*
		Caries free	0.927	0.936	0.907	(−0.021 [−0.075, 0.035])	(0.009, [−0.040, 0.058])	(−0.030, [−0.077, 0.016])
		Early caries	0.755	0.82	0.635	(−0.120, [−0.279, 0.038])	(0.066, [−0.060, 0.207])	(−0.186, [−0.315, −0.065])*
		Moderate-to-severe caries	0.939	0.902	0.826	(−0.113, [−0.193, −0.035])	(−0.037, [−0.096, 0.022])	(−0.076, [−0.167, 0.004])
Smooth Surface	Unimodal	Macro	0.687	0.737	0.806	(0.119, [−0.106, 0.288])	(0.051, [−0.084, 0.174])	(0.068 [−0.155, 0.234])
		Caries free	0.918	0.952	0.974	(0.055, [0.019, 0.094])*	(0.037, [−0.003 0.074])	(0.022, [−0.002, 0.046])
		Early caries	0.343	0.533	0.804	(0.461, [0.165, 0.756])*	(0.190, [−0.084, 0.488])	(0.271, [0.062, 0.479])
		Moderate-to-severe caries	0.8	0.727	0.64	(−0.159, [−0.809, 0.200])	(−0.072 [−0.303, 0.128])	(−0.087, [−0.757, 0.324])
	Multimodal	Macro	0.725	0.756	0.784	(0.059, [−0.161, 0.230])	(0.031, [−0.110, 0.164])	(0.028, [−0.190, 0.194])
		Caries free	0.929	0.954	0.97	(0.042, [0.006, 0.081])*	(0.026, [−0.011, 0.065])	(0.016, [−0.009, 0.041])
		Early caries	0.474	0.535	0.751	(0.278, [−0.018, 0.580])	(0.061, [−0.225, 0.370])	(0.216, [−0.027, 0.446])
		Moderate-to-severe caries	0.773	0.778	0.631	(−0.142, [−0.786, 0.235])	(0.005, [−0.227, 0.195])	(−0.148, [−0.788, 0.267])

* Denotes that the confidence interval of the difference crosses zero.

in children (primary, mixed, and permanent dentition). For the occlusal surface, the unimodal model showed similar macro-F1 scores across dentition stages (0.833-0.848), with no statistically significant differences between stages, indicating good cross-stage stability in occlusal caries detection. In contrast, the performance of the multimodal model varied significantly across dentition stages; it performed best in the mixed dentition stage (macro-F1= 0.886), while its performance in the permanent dentition stage was significantly lower than that in both the primary and mixed dentition stages, suggesting that its diagnostic performance is influenced by the dentition stage. For the smooth surface, both models exhibited a gradual increasing trend in macro-F1 scores from the primary dentition stage to the permanent dentition stage; however, the differences between stages did not reach statistical significance, indicating relatively stable overall performance.

SHAP analysis of the occlusal caries multimodal classification model revealed the top ten predictive features for each outcome class, as summarised in Figure 2A to C. Teeth classified as caries-free were typically associated with participants who reported no tooth pain or tooth discolouration during the past 6 months, a low impact of oral health on daily life, lower sweet food frequency, higher maternal education, more frequent flossing, belief that sugar causes caries, and a lack of awareness that fissure sealing can prevent caries. For the early caries class, the dominant features included recent bad breath or cavities, not being an only child, infrequent flossing, higher sweet-food intake, no history of dental visits, belief in the oral-systemic health link and in sugar as a cariogenic factor, together with a lack of awareness of pit and fissure sealants or of the bacterial role in caries. Participants whose teeth were predicted to have moderate to severe caries usually exhibited recent halitosis or tooth pain, not being an only child, predominant bottle-feeding during the first 6 months of life, frequent nighttime sweet consumption, not using toothpaste when brushing, belief in the oral-systemic health connection, and lack of awareness of the bacterial aetiology of caries.

The SHAP analysis for the smooth surface caries multimodal model revealed the top ten predictive features for each outcome category, as described in Figure 3A to C. Teeth classified as caries-free were typically associated with participants who had no dental visit history or whose most recent visit was primarily for treatment, along with low sweet-food frequency, gingival bleeding during brushing in the past 6 months, more frequent brushing, predominant breastfeeding in the first 6 months after birth, higher paternal education, the perception that oral health issues significantly affected eating, and lower oral health literacy. For the early caries class, key predictors included low overall sweet intake but frequent prebedtime sweet consumption, infrequent flossing, a recent dental visit mainly for treatment, food impaction in the past 6 months, the perception that oral health issues substantially affected household chores, and lower oral health literacy. Participants whose teeth were classified as moderate to severe caries usually exhibited infrequent brushing; the perception that oral health issues strongly impacted social interactions, tooth mobility and food impaction within the past 6 months; and lower levels of both oral health literacy and positive oral health attitudes.

Discussion

This study successfully established a multimodal model for caries screening by integrating intraoral image data with questionnaire-based metrics. Compared with the image-only unimodal model, the multimodal deep learning model increased early caries recall while preserving overall diagnostic performance. Furthermore, the model achieved an overall accuracy greater than 90%, fulfilling the study's primary objective and establishing its efficacy in automated caries screening.

Consistent with the findings of previous studies, the findings of this study confirm that AI has promising future applications in intelligent caries diagnosis. Li et al[40] developed a single model for the simultaneous detection of dental caries and periapical periodontitis, achieving F1 scores of 0.829 and 0.828, respectively. Its performance outperformed that of junior dentists and significantly improved the consistency and accuracy of dentists' diagnoses. A randomised controlled trial by Mertens et al[32] demonstrated that AI assistance could significantly increase the diagnostic sensitivity of dentists for dental caries, increasing it from 0.72 to 0.81. Zhang et al[13] evaluated an AI model integrating MobileNet-v3 and U-Net on 4361 teeth. The model achieved an overall accuracy of 93.40% and a high specificity of 95.65%. Despite this strong performance, its sensitivity varied considerably across tooth positions and caries types; for proximal caries, the sensitivity was notably low at approximately 10%. These findings suggest that these models are better suited as screening aids than standalone diagnostic tools.

The application of AI, particularly deep learning, for the automated detection of early caries presents significant challenges despite its promising potential. While several methodologies have been explored, the performance for initial lesions remains suboptimal. For instance, Jones et al[29] demonstrated that a deep learning model for segmenting caries on 3D intraoral scans achieved good performance in detecting extensive caries (sensitivity 71%, precision 66%), with limitations noted for early and moderate lesions. This difficulty in identifying early-stage caries was further highlighted by Moutselos et al,[30] who, using an R-CNN on a small dataset, reported high precision (0.78) for advanced lesions but markedly low recall (0.27) for ICDAS 2 lesions. Felsch et al[31] used a vision transformer for caries segmentation, achieving pixelwise IoU scores of 0.630 for noncavitated caries and 0.692 for dentine cavities. Despite the use of a large dataset (n = 18,179), the modest performance of the model for early lesions indicated that precise delineation remains challenging. Thus, the innovative aspect of our work stands out. In this study, questionnaire data were integrated to construct a multimodal model for enhancing the diagnostic sensitivity for early caries. While previous efforts have explored multimodal data for caries prediction, such as the system proposed by Njimbouom et al,[41] which combined survey-derived features with images using a hybrid neural network for binary classification on external datasets, the present study

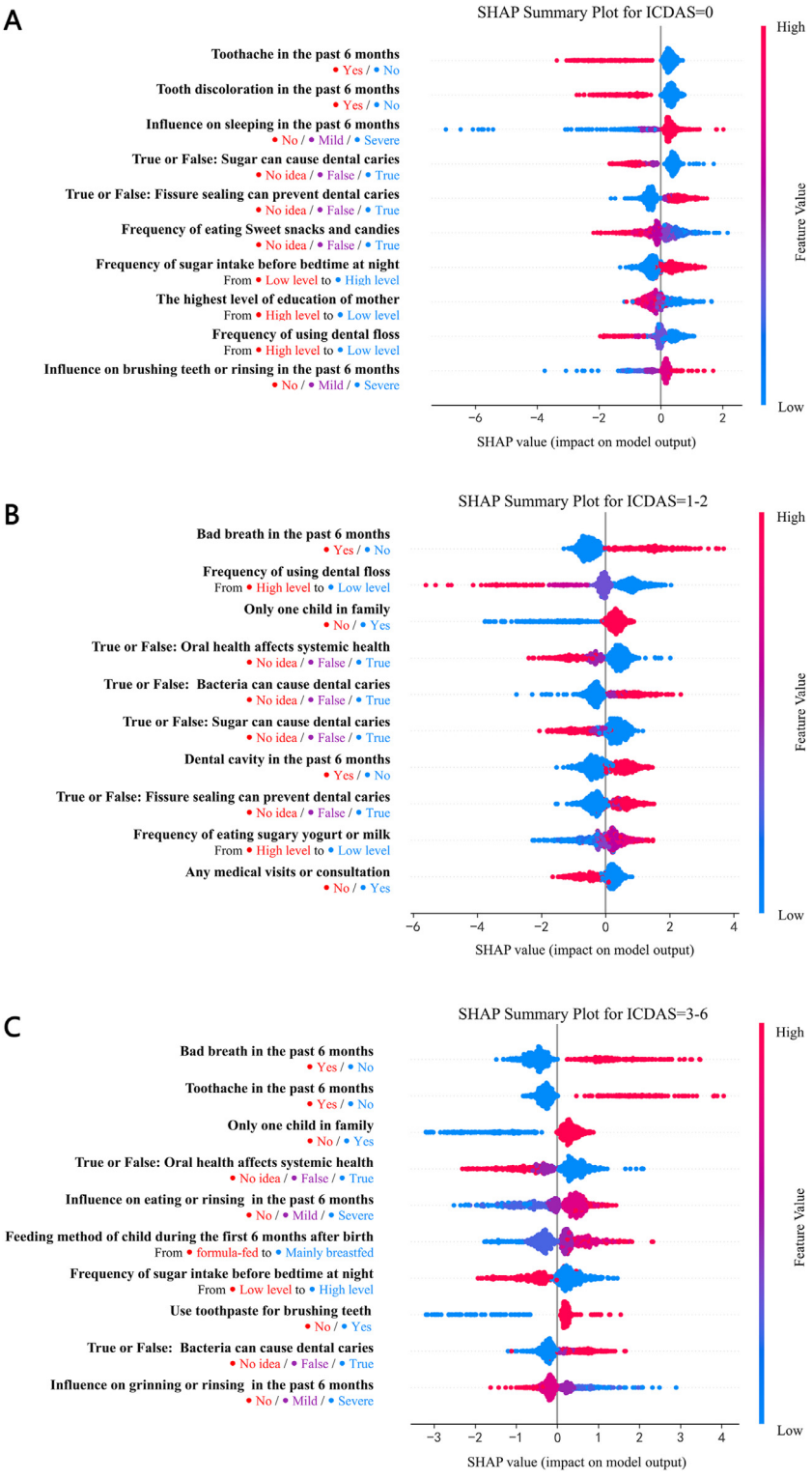


Fig. 2 – SHAP summary plots of each class in the caries classification model of the occlusal surface. A-C, represent the characteristics of caries-free, early caries, and moderate-to-severe caries on the occlusal surface. Y- axis: Features, which correspond to questions from the questionnaires, are ranked in descending order based on their mean absolute contribution to the model's output. X-axis: SHAP Value, which quantifies the feature's contribution to the model's prediction for a specific sample. A positive SHAP value indicates that the feature's specific answer increases the model's prediction output. A negative SHAP value indicates that the answer decreases the prediction output. Colour of dots: Feature value, which means the specific answer to the question. Individual dots: Each dot on the plot represents a single sample in the dataset. The horizontal density of the points reflects the range and concentration of SHAP values contributed by that feature value across different samples.

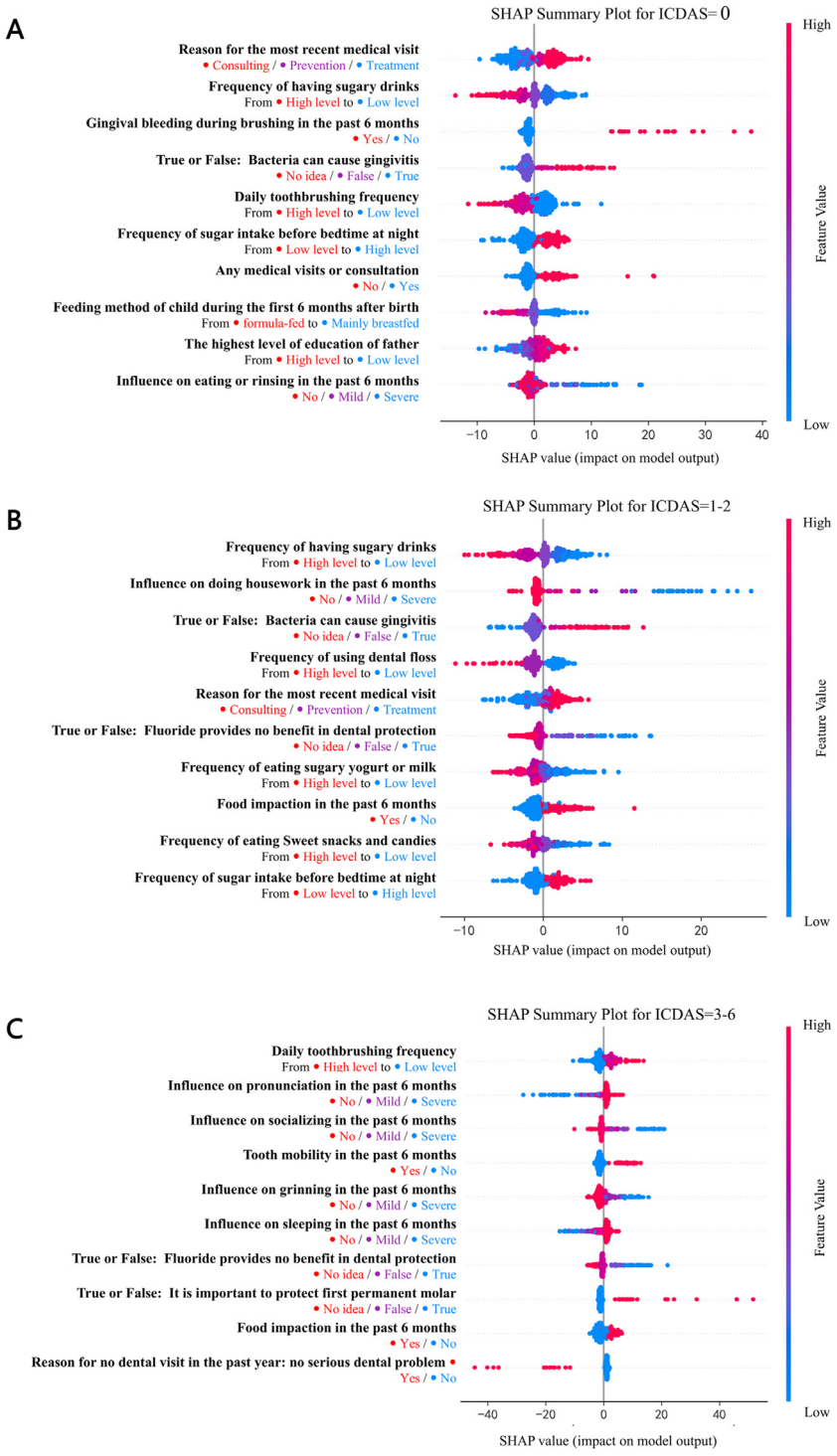


Fig. 3 – SHAP summary plots of each class in the caries classification model of the smooth surface. A-C, represent the characteristics of caries-free, early caries, and moderate-to-severe caries on the smooth surface. Y-axis: Features, which correspond to questions from the questionnaires, are ranked in descending order based on their mean absolute contribution to the model's output. X-axis: SHAP value, which quantifies the feature's contribution to the model's prediction for a specific sample. A positive SHAP value indicates that the feature's specific answer increases the model's prediction output. A negative SHAP value indicates that the answer decreases the prediction output. Colour of dots: Feature value, which means the specific answer to the question. Individual dots: Each dot on the plot represents a single sample in the dataset. The horizontal density of the points reflects the range and concentration of SHAP values contributed by that feature value across different samples.

advances the field by performing tooth-level, multicategory classification in a paediatric population within a real-world school setting. By enabling fine-grained, clinically relevant caries diagnosis within a multimodal framework, this work represents a meaningful advancement over existing approaches.

Current research regarding oral health monitoring across different dental stages (deciduous, mixed, and permanent dentition) remains limited.[43,44] This study used a standardised imaging device to capture dental images representing these distinct dental phases, creating a comprehensive multistage dataset to enhance model generalisability across paediatric age groups. Notably, our dataset comes from school populations rather than clinical conditions. Population-derived samples better reflect the natural distribution of dental conditions encountered in real-world screening environments, thereby improving the robustness of the model in practical applications. This design minimises selection bias while preventing artificial inflation of caries prevalence estimates, a common limitation of clinic-based training data that could compromise public health utility.[45] However, the population-based sampling approach inherently resulted in an imbalanced class distribution, with noncarious teeth predominating over early and moderate-to-severe caries. This disparity was particularly evident in smooth surface caries representation. To address these challenges, we implemented a multifaceted optimisation framework that incorporates data augmentation, L2 regularisation, and asymmetric loss functions to mitigate class imbalance effects.[46] This study used a dentition stage-stratified analysis to assess the performance of unimodal and multimodal models across different developmental stages in children. For occlusal surface detection, the unimodal model demonstrated consistent performance across all dentition stages. In contrast, the multimodal model performed significantly better in the mixed dentition stage than in the permanent dentition stage. This difference may be attributed to the ability of the multimodal model to integrate imaging data with individual-level features, potentially enhancing its sensitivity to developmental morphological variations. In the permanent dentition stage, where the occlusal surface morphology is more complex and exhibits greater interindividual variability, such performance variations may become more apparent. For smooth surface detection, both models maintained stable performance across all dentition stages.

Our findings indicate that both the unimodal (image-only) and multimodal models achieve excellent diagnostic performance, comparable to or exceeding that reported in previous studies.[11,43] The multimodal model significantly improves the recall rate for early caries detection while maintaining high overall classification accuracy. This represents a critical advancement, as early identification facilitates minimally invasive and cost-effective treatment strategies, as underscored by Schwendicke et al.[42] Notably, the questionnaires used were based on a standardised national oral health survey instrument. This suggests that features such as diet, dental hygiene, and dental visit habits are generally applicable risk indicators. However, cultural or regional differences in questionnaire responses may influence the model. Future versions of the model may require recalibration for different populations, or a simplified questionnaire focusing on the most influential questions identified through SHAP analysis could be used to enhance the generalisability and practical utility of the model.

Compared with the image-only unimodal model, the multimodal models exhibited slight variations, both increases and decreases, across certain performance metrics. These differences likely reflect the dual influence of questionnaire data on model outputs. Individuals at higher risk of caries often present with elevated DT/dt values, and their associated questionnaire features tend to correlate with a greater number of caries-affected tooth images. This alignment may enhance the model's ability to identify caries cases more accurately. In contrast, while a single-modal model operates on a one-to-one correspondence between each tooth image and its respective label, the multimodal framework introduces a one-to-many relationship wherein a single participant's questionnaire data are linked to multiple tooth images and labels. This structure may introduce complexity or noise into the decision-making process, potentially affecting classification consistency. In the future, more advanced fusion strategies or multimodal attention mechanisms should be explored to determine if alternative architectures (such as aggregating information across a patient's teeth or using different sequential models for processing questionnaire data) could further enhance performance or consistency in a multitooth context.

According to the data in Table 2, the precision for early caries on occlusal surfaces decreased from 0.792 to 0.727 with the multimodal model and from 0.621 to 0.583 on smooth surfaces. This indicates a slight increase in false positives for early lesions when the questionnaire data are used. In accordance with contemporary caries management principles, early lesions on smooth surfaces are managed with professionally applied fluoride or high-fluoride toothpaste (1.1% NaF), whereas early occlusal caries are treated with fluoride application or sealants, all of which are noninvasive and safe interventions.[47,48] Consequently, even if a healthy tooth is misclassified as having early caries (a false-positive) during screening, subsequent preventive measures such as fluoride therapy or sealant placement pose no harm to the child. Therefore, in screening, higher sensitivity at the expense of some specificity can be acceptable. However, it should be noted that the use of a questionnaire in screening programs means extra effort, as parents or children must complete the questionnaire. When real-world implementation is considered, evaluating whether the public health benefit gained from improved early caries detection is sufficient to justify the added step is essential. A feasible strategy to optimise the balance between effectiveness and resource use is a tiered screening approach: an image-only AI model could first be used for initial screening, with questionnaires administered only to those flagged or in borderline cases.

To enhance model interpretability, we used SHAP analysis to identify key features contributing to the classification of various types of dental caries within the multimodal model. This approach mitigates the 'black box' limitation inherent in artificial intelligence by quantifying feature contributions to caries severity predictions. The analysis revealed that children's dietary habits, oral hygiene practices, family

environment, health literacy, and the self-rated oral health-related quality of life were strongly associated with caries severity. These findings confirm and reinforce the importance of behavioural factors, such as dietary habits, oral hygiene practices, and parental knowledge, in the prediction of caries risk among children.[33,49] Notably, SHAP analysis revealed distinct feature dependency patterns between the occlusal and smooth surface caries diagnostic models. While oral health knowledge metrics predominantly drove occlusal surface classifications, smooth surface caries predictions were more strongly influenced by self-reported OHRQoL indicators. However, due to potential interactions among questionnaire features, caution is warranted when interpreting the individual effect of a single feature based solely on its SHAP value. Longitudinal or controlled experimental studies are needed to elucidate the causal mechanisms underlying these associations.

This research has several shortcomings. First, the diagnostic accuracy of clinical examiners represents a critical determinant of the reliability of caries classification. In our study, only one experienced dentist made all diagnostic decisions during clinical examination, which must be considered a potential source of bias. Second, the method of data collection was inconsistent. Oral health information for children aged under 12 years was obtained from parent reports, whereas participants aged 12 to 15 years completed self-administered questionnaires. This difference in data collection may introduce potential response bias and complicate the standardised interpretation of the results. Another limitation of this study is that the research is based on a single-centre cohort from Guangzhou, and the training-test split was conducted only at the patient level without stratification by key variables such as age or caries severity. This single-source sample, combined with the absence of a stratified design, may affect the fairness of the model evaluation and limit the generalisability of the results. Furthermore, all images were acquired using standardised intraoral photography equipment. While the uniformity in technical factors such as camera resolution and lighting conditions helps control internal variability, it also implies that the performance of the model in response to variations in equipment, imaging environments, or differences in caries prevalence across populations remains unclear. Therefore, although the current findings are promising, whether the model can truly serve as an effective population-level screening tool depends on its robustness in diverse real-world settings. This will require further validation through future studies involving multicenter populations with rigorous stratified sampling designs.

Conclusion

To our knowledge, this study presents the first multimodal model for screening for dental caries in children. Multimodal classification significantly increases early caries recall compared with the unimodal model while preserving overall diagnostic performance. Although artificial intelligence has shown great potential in clinical applications, comprehensive validation in larger-scale, multicenter studies remains necessary for achieving practical implementation.

Informed consent statement

Written informed consent was obtained from the patients and their guardians before the study.

Author contributions

Conceptualisation, methodology, formal analysis, investigation, writing – original draft, writing – review and editing: Liu, Pang.

Investigation, writing – review and editing: Li, Wang, Liu, Wang.

Conceptualisation, writing – review and editing: Lin.

Final approval of the version to be published, and agreement to be accountable for all aspects of the work: All authors.

Ethics statement

The study was conducted following the Declaration of Helsinki and approved by the Ethical Review Committee of Guanghua School of Stomatology, Sun Yat-Sen University (ID: KQEC-2023-87-01). This project has been registered with the China Clinical Trial Registry (ID: ChiCTR2400083125).

Data availability

Due to ethical considerations and the inclusion of participant-related privacy information, the clinical dataset used during this study is not publicly available. However, it can be obtained from the corresponding author upon reasonable request. Those interested in our software solutions should contact the author group.

Funding

This research was supported by the National Natural Science Foundation of China (grant no. 81903345).

Conflict of interest

None disclosed.

Acknowledgements

The authors sincerely thank all the participants and parents for their support and cooperation throughout the study. The authors are also very grateful to Y.J. Pang for his great support and contribution to this work.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.identj.2026.109420.

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